

Measuring the Wrong Thing: Internal Harmfulness Scores Anti-Rank Successful Jailbreaks

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Abstract

Internal safety scores judge a prompt before any text is generated, and they are validated by how well they separate harmful prompts from benign ones. That separation is then read as evidence that the score will also catch the attacks that succeed. Harmful intent is a property of the prompt. Jailbreak success is an outcome produced later by a particular target model, decoding policy, and judge. A filter tuned on a score that measures the wrong quantity spends its false positive budget on attacks that would have failed anyway. In this paper we audit that inference. Attention based measurements are usually read from prompt dependent locations, so a wrapper changes both the content being judged and the place the signal is taken from. We therefore introduce Active Attention Probing, which supplies a fixed content independent measurement coordinate. We pair every base goal with a plain and a wrapped version and generate real completions from the target models. On Llama, wrapping raises harmful generation from 0.05 to 0.27 while harmful intent AUROC falls from 0.936 to 0.803, so the attacks grow more dangerous while the prompts look safer to the score. Among wrapped harmful prompts the outcome AUROC is 0.220, which places the attacks that succeeded below the attacks that failed. Rare token, passive, and detector derived channels reproduce the reversal on the same matched design, and the reversal itself persists across three target models, seven attack families, and two independent judges. Distribution shift then degrades calibration and threshold transfer before it degrades ranking.

1 Introduction

An internal safety score can identify harmful intent before generation. A deployed filter may then treat a high score as evidence that the prompt will successfully jailbreak its target. These interpretations refer to different quantities. Harmful intent is a property of the prompt. Realized success also depends on the target model, decoding policy, and judge.

Our Llama results show that the two quantities can diverge. Wrapping a harmful request raises the harmful generation rate

from 0.05 to 0.27. At the same time, harmful intent AUROC falls from 0.936 to 0.803. Among wrapped harmful prompts, the score places successful attacks below failed attacks and reaches an outcome AUROC of 0.220. These results do not show that prompt safety detectors fail at their stated task. They show that harmfulness validation does not establish validity for realized success.

We test this inference with matched plain and wrapped versions of each base goal. We generate real completions from each target model and label the realized outcome. Attention based scores are usually read from prompt dependent locations, so a wrapper changes both the content being evaluated and the place the signal is taken from. A score change is then difficult to attribute to the model rather than to the moved coordinate. Active Attention Probing (AAP) removes this confound by providing a fixed content independent coordinate in the system region, so a prompt and its wrapped version are compared at a common reference point. AAP is the methodological contribution that makes the audit possible. The scientific claim of the paper concerns what the resulting score predicts. Rare token, passive, system span attention, and refusal logit channels test whether that claim depends on the AAP coordinate.

Figure 1 presents the audit. RQ1 asks whether a score validated on harmful intent can rank target specific jailbreak success. RQ2 then asks whether the original harmfulness interpretation survives distribution shift. We evaluate ranking, calibration, and threshold transfer separately because each can fail while the others remain useful.

Prior work shows that effective jailbreaks can suppress internal harmfulness features [5, 21]. Probes trained on success labels can also transfer poorly across attack families [17]. We ask what a score trained on harmfulness can validly support. We make three contributions.

- We distinguish prompt harmfulness from target specific jailbreak success and estimate both quantities on matched prompts with realized outcomes.
- We show that a strong harmfulness score can rank re-

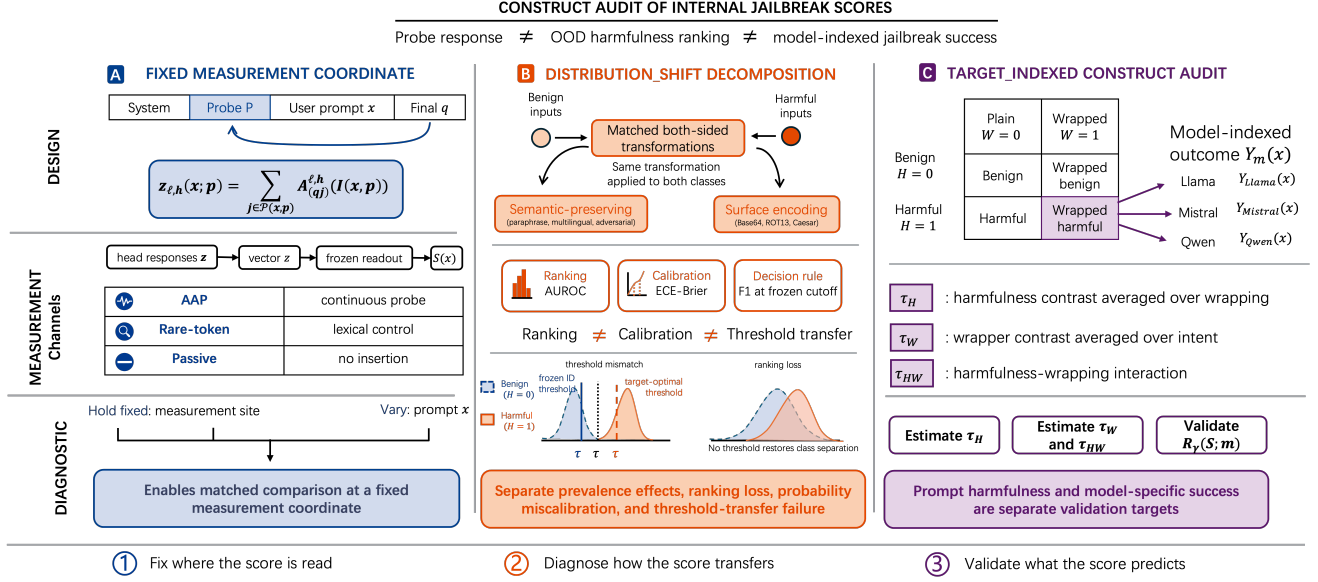


Figure 1: The audit reads one score against three questions. Each column is a question, and its rows give the design, the measurement channel, and the diagnostic. Panel A asks whether the response is measured at a stable place, so AAP fixes the coordinate and rare token and passive channels match it. Panel B asks whether the harmful intent reading survives a change of distribution, so every transformation is applied to both classes and ranking, calibration, and threshold transfer are separated. Panel C asks whether the score anticipates what the target does, so it crosses H with W over matched goals and validates against target indexed outcomes Y_m . Panel C is where the two constructs come apart.

alized success in the opposite direction. The result persists across measurement channels, targets, attacks, and independent judges. We also quantify the effect on a deployed filter.

- We test whether the original harmfulness interpretation survives distribution shift. The audit separates ranking loss, probability miscalibration, and threshold transfer.

2 Related Work

Internal safety scores. Attention Tracker and AttentionDefense use attention to detect prompt injection or adversarial prompts [12, 31]. HiddenDetect, GradSafe, refusal directions, mutation sensitivity, and deployed filters use logits, gradients, hidden states, prompt variants, or layered classifiers [4, 16, 28, 34–36]. This line of work establishes that internal signals discriminate prompt level safety or attack labels. We ask the separate question of whether such a score is a valid predictor of a realized, target indexed jailbreak outcome. We audit the interpretation supported by the existing validation and do not dispute performance on the stated prompt classification task.

Our audit treats each detector as a representation, a readout, and a threshold. A prompt level label validates harmful intent for any of these channels. It cannot establish whether the

target model will comply. GradSafe comes closest and compares gradients against a hypothetical compliant response that the target model never generates [34]. HiddenDetect reads a logit lens over multimodal inputs [16], and our HiddenDetect style row ports that readout to text only prompts with every deviation listed in the appendix. Attention head specialization under attack has been studied directly [35]. Our per head decomposition makes no specialization claim.

Jailbreak outcomes. Effective jailbreaks can suppress internal harmfulness representations and move harmful prompts toward a harmless region [5, 21]. Prompt side probes trained directly on success labels can transfer below chance across attack families [17]. That result uses Llama-3.1-8B, which is also our primary target. Our positive control asks a narrower question. It tests whether success information is available within template based wrappers over a shared goal set. It does not claim transfer across attack paradigms. Our main audit asks whether a score trained on harmfulness already has the orientation and operating meaning required for success prediction.

Jailbreak benchmarks report attack success for one generator and one judge. We retain both indices instead of treating success as a prompt property.

Validity under distribution shift. Safety detectors can degrade as attack styles and source distributions change [8, 18, 27, 29]. Single source splits also overstate safety detec-

tor AUC by 8.0 to 16.5 points [8]. That work proposes leave one dataset out evaluation. The leave one dataset out control removes corpus identity. We apply each transformation to both classes so that the transformation cannot serve as the label. Our both sided design removes transformation identity. No transformation experiment in that work addresses this shortcut. We also separate ranking, calibration, and threshold transfer. This decomposition identifies which part of the score interpretation fails.

AUROC is invariant to monotone score transformations while calibration and fixed thresholds are not. One fixed threshold metric therefore cannot support all three claims. That failure can be silent. Safety classifiers on frozen embeddings fall from 85% to chance ROC-AUC while mean confidence falls only 14% [29].

Probing as measurement. AAP uses a continuous embedding in the soft prompt and prefix tuning tradition [19, 20]. It provides a fixed measurement anchor instead of steering model behavior [37]. Passive, lexical, and norm matched controls test whether insertion, lexical content, or learning explains the result [10]. Response magnitude alone does not establish construct validity, so we validate every channel on held out labels. Every realized outcome is generated without the probe.

3 Measurement Setting and Estimands

Setting. We study a pre generation internal score computed from one eager attention prefill pass. The score has access to attention and hidden states. It sees the prompt x but not the realized completion.

Measurement targets. Let $H(x) \in \{0, 1\}$ denote harmful intent in the base request, $W(x) \in \{0, 1\}$ the presence of a jailbreak wrapper, and $Y_m(x) \in \{0, 1\}$ whether target model m actually emits a harmful completion under a frozen safety judge. We omit judge and decoding indices unless we vary them. An ineffective jailbreak has $W=H=1, Y_m=0$. An effective jailbreak has $W=H=Y_m=1$. Evidence that a score measures H establishes construct validity for harmful intent. Prediction of Y_m requires separate evidence. RQ1 audits that criterion claim. RQ2 tests whether validity for H survives a change in input distribution.

Validity criteria. Construct validity asks whether the score ranks its training label. Outcome validity asks whether it ranks Y_m for a named target and judge. Decision validity asks whether a threshold keeps its operating behavior. Evidence for an earlier claim does not imply a later one.

Deployment scope. We study a prompt side filter that blocks when the score exceeds a threshold. The relevant error includes successful attacks that score below failed attacks. A fixed benign false positive budget measures how the ranking allocates blocking capacity between the two groups.

4 Audit Design

Passive attention detectors aggregate locations determined by the prompt [12, 31]. A wrapper can therefore change both the input and the measurement location. AAP fixes this location. Let $x = (x_1, \dots, x_T)$ denote a tokenized chat prompt and let $p_\theta \in \mathbb{R}^{K \times d}$ denote a short probe embedding. $I(x, p_\theta)$ inserts the probe into the system region after the chat template is materialized. The user tokens remain unchanged. AAP reads the attention response at the probe and maps it to a scalar score. A matched passive read uses the same template boundary without inserting the probe.

Attention response. For layer ℓ , head h , probe positions $P(x, p)$, and final input position q , the response is

$$z_{\ell,h}(x, p) = \sum_{j \in P(x, p)} A_{qj}^{\ell,h}(I(x, p)). \quad (1)$$

The vector $\mathbf{z}(x, p)$ concatenates the monitored heads. XG-Boost maps this vector to a score $S(x) \in [0, 1]$. Rare token, common token, norm matched, system span, passive attention, and refusal logit channels test changes in probe content, insertion, location, and feature family. Response magnitude appears in the appendix.

Training and separation. The primary probe has $K=3$ rows. We train it only on the BeaverTails training split and freeze it before every audit. No JailbreakHub [30], WildJailbreak, or JailbreakBench prompt selects the probe, heads, or harmfulness readout. Every derived shift set starts from the held out test split. The harmfulness readout uses fixed constants and no validation search. The outcome supervised positive control selects regularization inside each cross validation fold. The appendix gives the optimization objective, initialization, head sensitivity, and continuous controls.

The probe starts from safety semantic token embeddings. The objective maximizes the safe versus harmful attention gap on the training split and uses only the training selected 20 head set. The dense RQ2 read and every held out domain stay unseen.

Three controls separate prompt surface, response magnitude, and transferred ranking. Head selection is the first. Sparse head subsets are high variance across random draws, so RQ2 uses the dense read. Prompt surface is the second. A length only classifier gives AUROC from 0.49 to 0.62, including settings where encoding makes prompts five times longer, so length does not carry the signal. Correlation between derived sets is the third. Source clustered intervals are 0.94 times as wide as unclustered intervals, so shared source prompts do not inflate significance.

Scores use `inputs_embeds` with the probe present. All generations use the original prompt and ordinary chat template without the probe. No reported outcome is produced by a probe perturbed generation. The appendix measures the insertion effect. Next token symmetric KL has median 0.22.

Refusal flips occur on 7.5% of prompts. A separate generation audit estimates a change in attack success of -0.03 with a 95% CI of $[-0.070, 0.007]$.

The appendix gives each control its read location, feature family, and readout. The safety semantic initialization alone does not reproduce the learned response. The gain comes from optimization and not from safety semantic content. These controls change one measurement choice at a time. They do not reimplement the published systems that motivated them. **Matched target design.** We cross harmful intent and wrapping while holding the base goal fixed. For $h, w \in \{0, 1\}$, define

$$\mu_{hw} = \mathbb{E}[S(x) \mid H(x) = h, W(x) = w]. \quad (2)$$

The balanced effects are

$$\begin{aligned} \tau_H &= \frac{1}{2}[(\mu_{10} - \mu_{00}) + (\mu_{11} - \mu_{01})], \\ \tau_W &= \frac{1}{2}[(\mu_{01} - \mu_{00}) + (\mu_{11} - \mu_{10})], \\ \tau_{HW} &= \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}. \end{aligned} \quad (3)$$

τ_H measures harmfulness, τ_W measures the paired wrapper effect, and τ_{HW} measures whether wrapping changes the harmfulness response. R_H denotes harmful intent AUROC. $R_Y(S, m)$ denotes outcome AUROC for effective and ineffective harmful prompts under target m . The same design applies to Llama, Mistral, Qwen, Gemma, and Phi.

The attention grids differ in layer and head count across these models. Equation 1 stays unchanged. Fixed score target substitution changes the generator while holding prompts, score source, decoding, and judge fixed. The target native matrix also changes the score source. This distinction separates generator effects from score architecture effects.

RQ1 uses the 20 heads selected on the training split. RQ2 uses the dense set of all heads. Both specifications read the same extracted attention tensor and both produce an outcome AUROC below chance. A benign quantile rule fixes the nominal false positive rate for the deployment audit [2, 24]. The appendix provides the finite sample bound, subset sensitivity, and the check that the deployed score recomputes from that tensor.

The rule uses the empirical $1 - \alpha$ score quantile of m safe calibration prompts. Sampling error and benign shift both loosen the guarantee. The audit therefore measures upper tail stability. It does not provide evidence about outcome validity.

5 Evaluation

Setup. Llama-3.1-8B-Instruct is the primary target. Mistral, Qwen, Gemma, and Phi test transfer across attention architectures. We curate BeaverTails [14] by removing duplicates and conflicting labels. The resulting pool contains 1774 safe and 3671 harmful prompts. A balanced set of 1000 prompts per class forms the training split. The held out split contains 774 safe and 2671 harmful prompts. We split source prompts before creating paraphrase, multilingual, or encoding variants.

Metrics. AUROC measures ranking. Brier score, ECE, and logistic recalibration slope measure probability calibration. Fixed and refitted F1 measure threshold transfer. Paired AUROC differences use DeLong tests and bootstrap 95% intervals. AUROC counts tied scores at one half. An ideal probability has a recalibration slope of one and an intercept of zero. We also report the F1 gap between a threshold frozen on the source domain and an oracle target threshold. The gap diagnoses the operating rule. It does not change the ranking score. The appendix reports every prespecified cell, the resampling units, and the complete statistical protocol.

5.1 Measurement Validation

Before auditing what the score predicts, we establish that AAP is a meaningful and controlled measurement channel. This subsection validates the framework both audit questions rely on and makes no claim about what the score predicts. The frozen harmfulness readout reaches a train to test AUROC of 0.985. Cross validation within the held out pool gives 0.957 to 0.982 across the five architectures. We treat the latter range as a separability check because each fold fits part of the held out pool. A paired intervention study also shows that learned probes increase response magnitude over lexical and norm matched controls without consistently improving ranking. We therefore treat AAP, rare token, passive, system span, and refusal logit reads as parallel measurement channels. The audit does not use response amplification as evidence of validity.

Norm matched controls confirm this distinction. Learned probes raise the response by at least sixteen times over a matched random probe on all three architectures, while linear readout AUROC moves by at most 0.06 and favors the random probe on Llama. Norm matching therefore removes embedding norm as the sole explanation for amplification. A larger response does not guarantee a better ranking.

5.2 RQ1: Does Harmfulness Validity Extend to Realized Success?

We pair the 100 benign and 100 harmful goals in Jailbreak-Bench [6] with plain and wrapped versions. Five wrapper families are balanced across intent classes. We freeze prompts, assignments, success criteria, and greedy decoding before inference. Llama Guard 3 provides the primary outcome labels [9, 13]. A frozen Qwen3-14B rubric relabels the cached completions.

The score measures harmful intent but reverses success. Figure 2 states the finding in one picture, and the paragraphs below give the estimates behind it. AAP has a harmfulness effect of $\tau_H = 0.521$ with CI $[0.458, 0.584]$, a wrapper effect of $\tau_W = -0.094$, and an interaction of $\tau_{HW} = -0.257$ with CI $[-0.359, -0.158]$. Wrapping selectively lowers the score for harmful requests. Harmful intent AUROC falls from 0.936 to

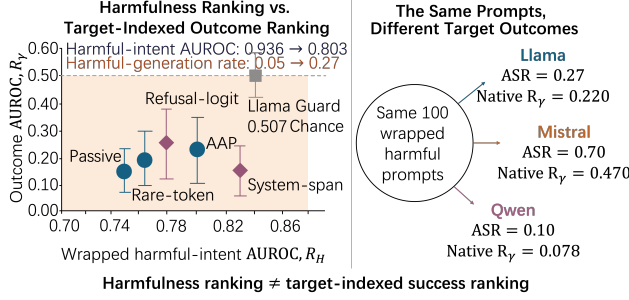


Figure 2: Harmful intent ranking and realized success ranking separate. The left panel shows harmful intent AUROC R_H and target outcome AUROC R_Y for each channel. Intervals are bootstrap 95% CIs and the dashed line marks chance. The right panel shows target specific success rates and outcome rankings for the same 100 wrapped harmful prompts.

0.803 while Llama harmful generation rises from 0.05 to 0.27. Among the 100 wrapped harmful goals, the score ranks the 27 successful attacks below the 73 failures. Outcome AUROC is 0.220 with CI [0.109, 0.343].

The induced outcome analysis reaches the same conclusion. Among 95 goals that Llama refuses without a wrapper, 23 become successful after wrapping. A McNemar test on the paired plain and wrapped outcomes gives $p = 3 \times 10^{-6}$. The score ranks these induced successes below the remaining failures with $R_Y = 0.254$ and CI [0.129, 0.396]. The plain harmful prompts also show an inverse ranking with $R_Y = 0.128$ and CI [0.025, 0.259]. The wrapper therefore does not create the reversal. What carries the signal is the wrapped score level rather than the plain to wrapped change, since ranking induced success by that paired difference gives an AUROC of only 0.421. A goal level logistic model keeps the coefficient negative after it controls the plain score, plain outcome, and wrapper family. Coefficients range from -0.59 to -1.33 across Llama and Qwen specifications. Within family centering gives $R_Y = 0.333$ with CI [0.236, 0.444]. Family stratified pooling, family fixed effects, and leave one family out estimates all agree. Successful attacks have the lower mean score in all five families. Three families contain only two successes, so we do not estimate a separate AUROC for each family. Five sampled decodes give R_Y from 0.202 to 0.252 under three outcome definitions. Every interval excludes chance and the greedy label is representative. The appendix reports the full induced, family, and decoding analyses.

The mismatch changes filter behavior. We set the threshold using benign prompts at a nominal 5% false positive rate. The achieved rate is 5.7%. The filter removes 42.5% of failed attacks but only 14.8% of successful attacks. The gap is -27.7 points with CI $[-44.8, -8.8]$. The filter admits 23 of the 27 realized jailbreaks. On the strong attack cohort, 92% of success-

ful attacks pass under either retained judge. Absolute harm still falls from 27 to 23, and the score removes about 2.6 times as many successful attacks as random blocking for the same benign budget. The score is not worse than no filter. It spends the budget mainly on attacks that already fail. The harmful rate among admitted prompts rises from 0.270 to 0.354. At a nominal 10% false positive rate the achieved rate is 11.1%. The filter blocks 80.8% of failed attacks and 29.6% of successful ones. The gap is -51.2 points. Preexisting misses explain 53% of leakage at the 5% point. The wrapper main effect explains 23%, and the inverse ranking explains 24%. The inverse share rises to 53% at the 10% point. An outcome supervised score reduces the admitted harm rate to 0.062 at matched coverage. Rare token and passive channels show the same allocation pattern.

Success information is available before generation. An outcome supervised readout uses the same internal features and grouped cross validation. Twenty five repeats keep every base goal within one fold. The readout reaches out of fold AUROC of 0.930 on wrapped harmful prompts and 0.939 on all harmful prompts. It reaches 0.875 with CI [0.800, 0.939] when the test wrapper family is held out and 0.725 with CI [0.680, 0.788] on WildJailbreak. Nested cross validation reproduces these values. A label permutation null gives 0.497 ± 0.072 . The best factorial baseline receives wrapper family success rates and still trails the readout by 0.156 on wrapped prompts. On WildJailbreak, a word level TF-IDF logistic model is the strongest baseline at 0.651 under the same protocol and regularization grid. The internal readout exceeds it by a paired margin from 0.016 to 0.139. These results show that the internal coordinates contain target specific success information. Qwen3 relabeling lowers the positive control ceiling to 0.883 but leaves the harmfulness score outcome AUROC at 0.220. Judge dependence limits the positive control. Qwen3 identifies 14 successful prompts, all within the 27 prompts identified by Llama Guard. Their agreement has $\kappa = 0.61$. The harmfulness score keeps the same outcome AUROC under both judges, but the Qwen3 labels cannot distinguish within family success from the family success rate. We therefore use the positive control to establish the presence of outcome information. We use Llama Guard alone for the stronger claim about the goal level component.

The result extends beyond AAP. Rare token, passive, system span, and refusal logit channels retain wrapped harmfulness AUROC from 0.751 to 0.828. Their outcome AUROC ranges from 0.145 to 0.248. Each channel also has a negative harmfulness by wrapper interaction and an outcome interval entirely below chance. The passive channel inserts no tokens, so probe insertion does not create either effect. Llama Guard remains a strong harmfulness discriminator and gives an outcome AUROC of 0.507. On 350 natural WildJailbreak prompts with 133 effective attacks, AAP outcome AUROC is 0.359. Llama Guard reaches 0.625 on the same cohort. That value exceeds every internal channel and its own matched de-

Score	Llama	Mistral	Qwen
Llama	.220 [.109,.343]	.491 [.363,.623]	.163 [.024,.358]
Mistral	.292 [.183,.411]	.470 [.345,.598]	.190 [.061,.343]
Qwen	.172 [.085,.275]	.386 [.272,.505]	.078 [.027,.139]
Wrapped success rate	.27	.70	.10

Table 1: Outcome AUROC R_Y for the same 100 wrapped harmful goals. Rows identify the native score and columns identify the generator. Entries include bootstrap 95% CIs.

sign value. Rare token, passive, and perplexity channels give 0.374, 0.449, and 0.487 [1]. The matched design remains the basis for the main claim. Across five probe seeds, wrapped harmfulness AUROC stays from 0.770 to 0.861 and outcome AUROC stays from 0.130 to 0.246. Per head analysis localizes the mismatch with Spearman $\rho = -0.85$ across 1024 heads. The appendix reports the mechanical null that limits this analysis to a descriptive role.

Targets change the outcome, not the conclusion. Table 1 recomputes the score on Llama, Mistral, and Qwen. Six of nine intervals lie below chance and none lies above chance. Mistral already complies with 0.68 of plain harmful requests, so wrapping barely raises its success rate. Its outcome AUROC near chance is therefore uninformative about an induced attack effect. Fourteen Mistral goals change from failure to success and twelve change in the opposite direction. McNemar testing gives $p = 0.85$. Five sampled decodes also show that target outcomes disagree at a rate of 0.81 against a no target effect null of 0.29. Pairwise native score correlations are only 0.42 to 0.54. Native recomputation does not recover a positive outcome ranking.

Strong attacks and independent judges reproduce the pattern. We evaluate GCG, AutoDAN, AutoDAN-Turbo, PAIR, TAP, Best-of-N, and Many-shot [3, 7, 11, 22, 23, 26, 38]. Table 2 reports Llama Guard 3, Qwen3, and a StrongREJECT rubric [32]. We exclude the StrongREJECT column from inference because the available substitute marks many verbatim refusals as successful. That leaves fourteen attack cells, which with the nine target cells of Table 1 make twenty three in total. Six of the nine and six of the fourteen have intervals below chance, and no interval lies above chance. A bootstrap clustered by goal and attack family gives a pooled outcome AUROC of 0.313 with CI [0.212, 0.422] and $p = 0.0006$. Those twenty three cells share one pool of 200 base goals, and the same clustered bootstrap gives a design effect of 9.3, or about 2.5 independent units, so we report no sign test over cells. Qwen3 is independent of the search procedure for GCG, AutoDAN, PAIR, and TAP. GCG and AutoDAN remain below chance under Qwen3 after BH FDR correction. Pooled logistic models also control the plain goal score and attack family. They retain negative score coefficients of -0.54 under Llama

A. Attack success rate				
Attack	n	LG3	SR	Q3
Best-of-N	200	.10→.94	.30→.73	.04→.26
Many-shot	200	.10→.18	.32→.30	.03→.18
GCG	100	.11→.17	.28→.25	.02→.09
AutoDAN	100	.05→.34	.29→.49	.03→.20
PAIR	100	.05→.52	.26→.78	.01→.07
TAP	50	.02→.16	.20→.46	.00→.06
AutoDAN-Turbo	50	.02→.38	.22→.51	.00→.14

B. Outcome AUROC R_Y				
Attack	n	LG3	SR	Q3
Best-of-N	200	.35	.43	.54
Many-shot	200	.48	.53	.49
GCG	100	.19	.27	.26
AutoDAN	100	.20	.14	.24
PAIR	100	.48	.38	.43
TAP	50	.31	.25	.21
AutoDAN-Turbo	50	.28	.25	.51

Table 2: Strong attack results on Llama. Panel A reports plain to attacked success rates. Panel B reports outcome AUROC. LG3 and Q3 are Llama Guard 3 and Qwen3. SR is shown for transparency and excluded from inference.

Guard and -0.35 under Qwen3. Base goal clustered intervals exclude zero for both judges.

Every attack row pairs the plain and attacked version of the same goal under the frozen Llama scorer. The $n = 200$ rows combine JailbreakBench and HarmBench [25]. The StrongREJECT substitute marks about 30% of verbatim refusals as successful and also fires on benign controls. The inverse ranking survives strict consensus and union labels in the appendix. We found a checkpoint defect in our own pipeline and regenerated five search based rows. No conclusion changes. The appendix reports the repair and every interval.

Qwen3 agrees with Llama Guard on 0.87 of wrapped Llama outcomes and on only 0.66 for Mistral. It preserves the AAP, rare token, and passive inverse rankings on Llama. These results show that one judge does not create the matched reversal. They do not remove the broader uncertainty from automated outcome labels.

Attention dilution does not explain the reversal. Fixing the probe position does not fix its share of the softmax, and attacks can target that share directly [33]. Wrapping lowers total attention to the system region from 0.383 to 0.225. Dividing by system region attention is a clean dilution control because that region holds a constant 13 tokens and the wrapper lands in the user turn, so the denominator is a fixed competition set. The outcome ranking remains inverse after that normalization and after normalizing by the full attention row. Sequence length is not the driver either, since prompt length explains only 4% of score variance. The appendix reports both normalizations, the retrained normalized readout, and the probe perturbation audit.

Together, these results are consistent with a representa-

tional decoupling account. Wrapping may weaken features used by a readout trained on harmful intent while separately changing the model processes that govern compliance. The passive and refusal logit results argue against probe insertion as the source, and the outcome supervised readout shows that success information remains available. We treat this account as a hypothesis because the audit does not identify a unique causal pathway.

RQ1 finding. A score that ranks harmful intent well can rank realized jailbreak success in the opposite direction. Harmfulness validation does not establish validity for success prediction.

5.3 RQ2: Does Harmfulness Validity Survive Distribution Shift?

RQ1 shows that validating a score against one construct does not license reading it as a predictor of another. We now ask whether the score stays valid for the construct it does measure once the input distribution changes. A shift can break that reading in three ways. The ranking of harmful above benign can fail, the numerical meaning can fail while the ranking holds, or only the transfer of a previously chosen threshold can fail while both hold. A single fixed threshold number cannot say which occurred.

We therefore keep the construct fixed as harmful intent and vary the input distribution. Meaning preserving shifts include WildJailbreak, paraphrases, and English, Chinese, French, and Japanese translations [15]. Surface transformations include base64, rot13, and caesar encoding. We apply every transformation to both classes. This design prevents the transformation from serving as the label.

Readout family is a material source of variation. Logistic regression and a linear SVM match the source AUROC near 0.98. Their mean AUROC across the same shifted domains falls to 0.55 and 0.64, compared with 0.75 for XGBoost. On the Japanese condition, the linear readouts reach 0.01 to 0.08 and XGBoost reaches 0.228. We keep XGBoost fixed as the primary readout and give the analysis in the appendix.

The Japanese result is an inversion with a known transformation artifact. Translated prompts have 2.4 times the character trigram similarity of the English source text. The Japanese transformation therefore collapses prompts toward a shared register. Its orientation free AUROC is $1 - 0.228 = 0.772$, close to the Chinese and French range. We retain the row as a readout sensitivity case and do not treat it as clean evidence of semantic robustness failure.

Thresholds fail before rankings. Table 3 shows that the frozen decision rule loses more F1 than the score loses AUROC on meaning preserving shifts. The source readout is not a calibrated probability. Its slope is 0.545. The slope column is an in sample Cox fit. Platt scaling on a frozen source slice changes the slope to 1.019 and ECE from 0.045 to 0.009. The calibrated source rule reaches WildJailbreak F1 of 0.667 without target labels. The oracle reaches 0.678. The gap re-

Domain	AUROC	ECE	slope	F1@0.5	F1@ $\tau_{20\%}$
In distribution	0.985	0.045	0.54	0.925	0.937
WildJailbreak	0.850	0.366	0.51	0.559	0.664
Paraphrase	0.858	0.203	0.32	0.734	0.769
English round trip	0.987	0.041	0.76	0.950	0.950
Multilingual (zh)	0.802	0.247	0.43	0.731	0.719
Multilingual (fr)	0.778	0.232	0.42	0.688	0.749

Table 3: Ranking, calibration, and threshold transfer under meaning preserving shifts. F1@0.5 freezes the source threshold. F1@ $\tau_{20\%}$ relocates only the threshold using a labeled target slice.

mains from 0.006 to 0.020 on domains where AUROC stays from 0.76 to 0.86. This repair is not universal. On the English round trip control, source calibration changes ECE from 0.041 to 0.171 and slope from 0.761 to 1.488. The frozen calibration map can therefore harm a domain that already ranks well. The appendix reports the remaining calibration checks.

The appendix separates a score that was never calibrated from a score whose calibration changes under shift. The source map repairs the source domain and improves five shifted domains. It does not restore a probability interpretation everywhere. The encoding slopes remain near zero because the ranking has disappeared. The Japanese slope remains negative because the class ordering has reversed. No monotone calibration map can repair either case. The ranking loss also replicates across architectures. Mistral, Qwen, Gemma, and Phi repeat the WildJailbreak evaluation and their AUROC falls to a range of 0.60 to 0.85.

Prevalence explains only part of threshold shift. We reweight each domain to a common balanced class prior. AUROC is unchanged. The source common prior threshold reaches WildJailbreak F1 of 0.790, compared with an oracle value of 0.796. HiddenDetect and refusal direction scores keep larger threshold gaps under the same audit, while the AttentionDefense style channel is better calibrated than AAP on WildJailbreak. Every score family loses separation on both class base64. A benign only quantile rule holds its nominal false positive rate in distribution and nearly matches oracle recall on paraphrase and English prompts. Under benign shift its WildJailbreak recall falls to 0.03 against an oracle of 0.25. This identifies benign tail stability as a separate requirement. The appendix reports the per score common prior table.

Both class encoding removes harmfulness separation. Encoding only harmful prompts gives AUROC from 0.85 to 0.89. A model free character entropy statistic reaches AUROC 0.997 to 1.000 on that single class comparison and 0.513 on unencoded prompts. Encoding both classes gives 0.51, 0.50, and 0.52 for base64, rot13, and caesar. Mistral, Qwen, Gemma, and Phi reproduce a mean AUROC from 0.50 to 0.51. The single class design measures a transformation cue. The both class design shows that the score loses harm-

fulness separation when the cue is unavailable.

RQ2 finding. Distribution shift can change calibration and threshold transfer while ranking survives. Stronger surface shifts can also remove the harmfulness ranking itself.

6 Discussion and Limitations

Implications. A benchmark should name the construct attached to each label: harmfulness detection requires H and R_H , whereas realized success requires generations and Y_m indexed by target, decoding policy, and judge. A complete audit should report harmful-intent ranking on held-out prompts, realized outcomes for protected targets, and blocking at a fixed benign budget. The evaluated thresholds reduce absolute harm but allocate blocks poorly within the harmful cohort, assigning lower scores to successful attacks; low harmfulness therefore does not imply refusal.

Limitations. Outcome labels are automated, without a blinded human audit, and three-judge agreement is low (Fleiss $\kappa \approx 0.3$); judge choice thus changes Y_m but does not itself support our thesis. Llama Guard guides several attack searches, leaving only the rubric judges independent for those rows. Primary outcomes use one greedy decode; five sampled decodes preserve the factorial result, whereas the strong-attack and shift results remain single-draw. The Llama and Qwen cells contain only 27 and 10 positives, and the targets are not difficulty-matched, so they show a range of behavior rather than a controlled comparison. Strong attacks and derived channels are evaluated on Llama; only Best-of-N and Many-shot were repeated on Mistral and Qwen. Probe insertion perturbs the prefill, although outcomes use unmodified prompts. The Japanese transform contains an artifact, and the lowest OOD results depend on readout family.

Scope and artifacts. The estimates cover only the tested prompts and checkpoints, so new deployments require new outcome audits. The matched factorial design supports the main claim. WildJailbreak and strong attacks are uncontrolled external checks. We release an artifact containing the analysis code, configurations, tests, and publication-aligned numeric aggregates at <https://github.com/kuangren777/aap-audit>.

7 Conclusion

A strong harmful intent AUROC does not establish that a score can predict jailbreak success. Our matched audit shows that the score can place successful attacks below failed attacks. The result persists across measurement channels, targets, attack families, and independent judges. Distribution shift can also change the original harmfulness interpretation. Harmful intent and realized success require separate validation.

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A Additional Ablations

These ablations support the measurement and evaluation choices in the main paper.

Probe optimization. The learned $K=3$ continuous probe is initialized from safety-semantic token embeddings and optimized only on the BeaverTails training split with Adam, learning rate 10^{-3} , 50 optimizer updates, each on a freshly sampled mini-batch of 20 prompts per class. The objective is

$$\mathcal{L}(p) = -\frac{1}{|\mathcal{H}|K} \sum_{h,k} \left| \bar{z}_{h,k}^{\text{safe}}(p) - \bar{z}_{h,k}^{\text{attack}}(p) \right| + 0.1 \bar{z}^{\text{safe}}(p) + 0.01 \|p\|_{\text{row}}. \quad (4)$$

The first term maximizes the absolute source-domain attention gap, the second discourages high safe-prompt attention, and the last controls row-wise embedding norm. No OOD or target-audit prompt is used for probe or readout selection.

Benign-only quantile rule. For trusted safe calibration prompts $C_0 = \{x_i\}_{i=1}^m$ and false-positive budget α , we set

$$\hat{\tau}_\alpha = \hat{Q}_{1-\alpha}(\{s_\theta(x_i; p) : x_i \in C_0\}), \quad (5)$$

$$\hat{y}(x) = \mathbb{I}[s_\theta(x; p) > \hat{\tau}_\alpha].$$

The Dvoretzky–Kiefer–Wolfowitz bound gives sampling error $\sqrt{\log(2/\delta)/(2m)}$. Benign shift adds $\gamma_d = \sup_t |F_{0,d}(t) - F_{0,\text{cal}}(t)|$. The diagnostic therefore identifies upper-tail stability rather than outcome validity.

Head-coverage sensitivity. Sparse 20-head subsets are high-variance. Across 40 random 20-head draws the minimum AUROC over nine OOD domains is 0.60 ± 0.15 (95% CI $[0.25, 0.83]$), and the layer-diverse and safety-ranked heuristics fall at the 65th and 47th percentiles of that distribution, inside the noise. We therefore report the all-head read as a fixed measurement specification and treat subset choice as a sensitivity analysis.

Readout sensitivity. The in-distribution result is similar for XGBoost, logistic regression, and linear SVM (5-fold AUROC ≈ 0.98). Under shift, however, the dense all-head result depends on the readout family: linear probes on the full feature degrade far more than the tree readout on the lowest-ranked both-sided domains (Japanese: 0.01–0.08 against XGBoost’s 0.228). We therefore report XGBoost as the fixed primary readout and make this dependence explicit.

Surface-form and correlation controls. Prompt length does not drive the signal: a length-only classifier is near chance everywhere (0.49–0.62 AUROC), including on encoding where attacks are $5\times$ longer. We recompute confidence intervals with a bootstrap clustered by source prompt, because the encoding, paraphrase and multilingual sets share source prompts. The clustered interval is $0.94\times$ as wide as the unclustered interval (a 6% narrowing), so within-source correlation does not inflate our significance. Finally, exploratory retraining shows that increasing the in-distribution attention gap can reduce OOD AUROC toward chance. Response magnitude and transferred ranking are therefore empirically distinct.

Norm-matched continuous controls. We construct a random three-vector probe with each row normalized to the corresponding learned row norm. It therefore has the same length,

insertion coordinate, dtype, and per-row embedding norm as the learned probe. On 100 + 100 BeaverTails prompts, the Llama learned/random controls give response magnitude 1.008/0.046 and linear-readout AUROC 0.931/0.944. Mistral gives 2.054/0.124 and 0.964/0.906. Qwen gives 0.802/0.030 and 0.932/0.896. The control rules out embedding norm as the sole source of amplification, while the differing ranking outcomes illustrate the architecture-dependent amplification–ranking interaction. Qwen’s non-finite entries affect fewer than 0.2% of responses and are excluded only from aggregate estimation.

Relative-anchor feature. We also tested a relative feature $r_{\ell,h} = \log((A_{\ell,h}^{\text{probe}} + \epsilon)/(A_{\ell,h}^{\text{ctrl}} + \epsilon))$ that normalizes probe attention by the matched natural-control read at the same head, intended to cancel sequence-length and global-attention drift across domains. On WildJailbreak (with per-prompt aligned probe/control features), the relative feature alone *underperforms* the raw probe read (AUROC 0.744 vs 0.803), while concatenating raw and relative features gives only a marginal gain (0.816). The relative feature varies across domains and the concatenated representation changes WildJailbreak AUROC only marginally. The reported analysis therefore uses the pre-specified raw attention read.

Factorial intervention details. The nine paired conditions are a natural three-token system control. Rare probes with 1, 3, and 5 IDs. A shuffled rare probe. Independently sampled rare IDs. Common IDs. Safety-semantic IDs, and the default rare triplet inserted at the user midpoint. Linear probes on the full head-response vector remain strong in distribution (mean five-fold AUROC 0.95–0.99 across the three displayed conditions and models), even when the scalar mean response has the opposite orientation. This confirms that the information is multivariate and cautions against interpreting the sign of an all-head average as classifier performance. Qwen produced non-finite values for fewer than 0.11% of head responses in these runs. Raw arrays were retained, while only summary estimation replaced non-finite entries by zero. Substituting exclusion or median imputation for the zero fill changes the reported statistics by less than their quoted precision, since the affected fraction is under 0.2%.

Hybrid representation. A hybrid variant concatenates the final-position projection onto a refusal direction and the hidden-state norm to the attention response. It is not used for any RQ1 or RQ2 estimate in the main paper. We report it only because the in-distribution measurement-validation numbers quote both variants, where it tracks the attention-only readout to within 0.005 AUROC on every architecture.

Strong-attack budgets and goal subsets. The seven attacks do not share one goal list, which is why their unattacked success rates differ. The $n = 200$ rows (Best-of-N, Many-shot) pool JailbreakBench and HarmBench harmful goals. GCG, AutoDAN and PAIR use 100 goals each and TAP and AutoDAN-Turbo use 50, drawn from the same pool but not identical subsets, so the plain-prompt rate in Panel A varies

by row (for example 0.11 for GCG against 0.05 for AutoDAN under Llama Guard) and rows should be compared to their own plain column rather than to one another. Search budgets were fixed before running: GCG uses 40 optimization rounds with early stopping once the target negative log-likelihood falls below 0.10. The tree and lifelong-memory searches (TAP, AutoDAN-Turbo) run at depth, width and branching factor 2 because of their cost. Best-of-N and Many-shot use their published sampling and shot budgets. All greedy attacks decode 256 new tokens, while query and sampling attacks retain their search-time winning completion.

Goal-difficulty control. The pooled model quoted in the main text is, for each judge,

$$\text{logitPr}(Y_i = 1) = \beta_0 + \beta_S \tilde{S}_i + \beta_P \tilde{S}_i^{\text{plain}} + \gamma_{a(i)}, \quad (6)$$

where $\tilde{S}_i$ is the standardized score of the attacked prompt, $\tilde{S}_i^{\text{plain}}$ the standardized score of the same base goal unattacked, and γ_a an attack-family effect. Intervals come from a bootstrap that resamples base goals as clusters.

B A Checkpoint Defect and Its Repair

While auditing the strong-attack table we found, and then fixed, a real defect in our own pipeline. We report it in full because it changes five rows of that table and because the failure mode is easy to reproduce in any resumable evaluation harness.

Mechanism. Both the scoring and the generation stages resumed from a checkpoint that was validated by *length only*, restarting at `len(cached)` in a prompt list rebuilt from a mutable manifest. Each attack script also has a smoke-test truncation. A pilot run over s goals produced a short manifest and complete checkpoints. The subsequent full run overwrote the manifest with B goals and silently reused the stale checkpoints as a prefix, displacing records by exactly $k = 2(B - s)$ positions. The displaced block lands on the wrong goals, because the records are laid out with the benign class first.

Proof and scope. Per-prompt scores are computed one prompt at a time on a frozen detector, so records sharing a base goal must be byte-equal across cells. Under the fitted k the displaced value equals the consensus value at position $i + k$ for every disagreeing record, on all five stored artifacts and in all five affected cells, and the fitted k matches $2(B - s)$ in each. Five of twelve cells are affected, namely GCG, AutoDAN, AutoDAN-Turbo, PAIR and TAP, all on the Llama target. Best-of-N and Many-shot are clean on all three targets. The attack stage itself is immune because its checkpoint is keyed by goal identifier, so no attacked prompt needed to be recomputed. The defect touched 240 completions and 320 scores. The score stage is corrupted for more records than the completion stage because scoring runs over all records while generation runs only over those still missing a completion.

Effect on the results. We regenerated the affected cells with an assignment bound to the record identifier, preserving each attack’s original completion semantics: GCG and AutoDAN are greedy throughout, whereas the query attacks retain their search-time winning completion, whose identity binding we re-verified against the attack checkpoint (400/400 matched on both prompt and completion). Ten of the fifteen repaired attack-by-judge rows are bit-identical to the published ones, because most of the displaced records fell in the benign half of the record list, which neither ASR nor R_Y reads. The five that move are the three GCG rows and the rubric entries for AutoDAN and AutoDAN-Turbo. GCG moves and strengthens: R_Y becomes 0.189/0.264 under Llama Guard and Qwen3, and its Qwen3 interval now excludes chance where it previously did not. Among the fourteen retained attack-by-judge cells the count significantly below chance is unchanged at six, with GCG–Qwen3 entering and TAP–Llama Guard leaving. No conclusion in the paper changes.

C Per-Head Decomposition and Its Mechanical Null

Per-head decomposition. We re-extracted the per-head probe response for the locked 400-prompt factorial cohort as a dense tensor of shape $400 \times 32 \times 32 \times 3$ (prompt $\times$ layer $\times$ head $\times$ probe position). Recomputing the deployed scalar score from this tensor reproduces the cached per-prompt scores bitwise (Pearson $r = 1.000$, maximum absolute deviation 0 over all 400 prompts), so the per-head features come from exactly the computation that produced every reported number. Three independent extractions were byte-identical.

Scoring each of the 1024 heads on its own, per-head harmful-intent AUROC and per-head outcome AUROC vary widely, and the two are strongly anti-correlated (Pearson -0.845 , Spearman -0.849). The five heads that rank intent best average $R_Y = 0.117$ and the twenty best average $R_Y = 0.121$. We do not quote any individual head’s value: with 27 positives, a maximum taken over 1024 heads is not interpretable in sample.

This anti-correlation is mostly mechanical, and we say so rather than reading it as functional specialization. Any head that ranks intent well would anti-rank outcome with no specialization involved at all, were the outcome a deterministic threshold on the aggregate score. Simulating exactly that null, matched to the observed positive count and to the observed aggregate R_Y , already produces a mean head-level Spearman correlation of -0.71 (95% range $[-0.83, -0.48]$) along the published score direction, and -0.82 along an alternative aggregate direction, where the null range covers our observed value outright. Our observed -0.849 sits outside the null band for the published direction ($p = 0.006$) and the partial correlation controlling each head’s loading on the aggregate direction is -0.59 , also outside its null band, so a residual

component survives, but the effect is not robust to the choice of aggregate direction, and we therefore report the decomposition as a localization of where the two criteria diverge rather than as evidence that distinct heads encode distinct constructs. By layer band, early heads (layers 0–10) are at chance on both criteria (R_H 0.510, R_Y 0.510), while middle (11–21) and late (22–31) heads discriminate intent and anti-rank outcome (0.708/0.267 and 0.724/0.260 as head means, 0.098 and 0.104 when the band is aggregated into a single score). Outcome-informative heads nevertheless exist: 266 heads have $R_Y > 0.5$ in sample and 22 survive a max-statistic permutation correction at the family-wise level, against roughly five expected under the matched null. The dense all-head read gives $R_H = 0.882$ with $R_Y = 0.087$, and the deployed twenty-head read gives 0.803 with 0.220. The qualitative conclusion is the same under both.

Wrapper-family stratification. Three of the five wrapper families produce only two successes each, so no per-family outcome AUROC is estimable and the marginal estimate is dominated by families with more events. Within-family centering gives $R_Y = 0.333$ (CI [0.236, 0.444], bootstrap $p = 0.003$, permutation $p = 0.011$) and Mantel–Haenszel stratified pooling gives 0.323 (CI [0.178, 0.472], $p = 0.020$). Leave-one-family-out estimates range over 0.296–0.345. Only 15.4% of the 1971 discordant pairs are within-family, which is why the stratified estimate is the less precise of the two. Restricting to the two adequately powered families gives 0.313 with an interval that just touches chance ([0.144, 0.503]). Between families, mean score and attack success are negatively related but not significantly so with five points (Pearson -0.77 , $p = 0.13$).

We deliberately do not rest homogeneity on a non-significant interaction test. A power simulation calibrated to the observed family intercepts shows that the score-by-family interaction likelihood-ratio test on these data has a rejection rate of ≈ 0.16 both under a true null and under an alternative in which one family’s slope is fully sign-flipped: its power equals its type-I rate, so a non-rejection carries no information. The homogeneity evidence we do report is sign consistency: in all five families the successful attacks have the lower mean score (differences -0.34 , -0.36 , -0.31 , -0.10 , -0.03).

Outcome-supervised readout. Training on the outcome label with grouped repeated cross-validation (25 repeats, base goal never spanning folds) gives out-of-fold AUROC 0.930 on the wrapped harmful cohort, 0.939 on all harmful prompts, and 0.725 (CI [0.680, 0.788]) on the 350-prompt WildJailbreak audit, against label-permutation nulls of 0.50 ± 0.07 ($p = 0.005$ in each case). Leave-one-wrapper-family-out gives 0.875 (CI [0.800, 0.939]). Nested cross-validation reproduces these values, so selecting the regularization strength is not the source of the advantage. We quote bootstrap intervals rather than the spread across cross-validation repeats, which is not a confidence interval because the folds are not independent.

Every baseline is refitted under the readout’s own protocol and its own regularization grid, and the two cohorts must be read separately. On the JailbreakBench factorial cohorts the baselines span 0.62–0.80. The strongest is not a text model but an oracle that is told each wrapper family’s empirical success rate, and the readout still leads it by +0.156 (wrapped cohort) and +0.134 (all harmful). On WildJailbreak neither a wrapper-identity nor a plain-goal baseline exists, since those prompts are naturally occurring adversarial phrasings with no synthetic family and no matched unwrapped counterpart. There the baselines reach only 0.48–0.65, the best being a word-level TF-IDF logistic model at 0.651, a difference of 0.074 whose paired estimate is +0.077 (CI [+0.016, +0.139], $p = 0.007$). An earlier draft quoted a 0.087 margin on this cohort. That figure came from an inconsistent protocol in which the baselines were left untuned, and is superseded by the paired estimate above.

This result is judge-dependent in a way the main text notes and we detail here. Relabelling the same cohort with the frozen Qwen3 rubric leaves the inverse ranking essentially unchanged ($R_Y = 0.2201$ versus 0.2202 under Llama Guard), but that rubric records only 14 successes, a strict subset of Llama Guard’s 27 with $\kappa = 0.61$, and under it the supervised ceiling falls to 0.883 with the within-family component no longer resolvable. The claim that outcome information is present survives both judges. The claim that it is goal-level rather than wrapper-level is established only under Llama Guard labels.

D Deployment Risk and Normalization Controls

Operating points. We calibrate the block threshold on benign prompts only, splitting the held-out in-distribution safe pool in half so that calibration and verification are disjoint, and we report the achieved false-positive rate on the verification half (5.7% at a nominal 5%, 11.1% at a nominal 10%). At nominal rates of 0.1% and 1% the threshold sits above every score in this cohort and nothing is blocked. That is a score-scale shift between the calibration corpus and JailbreakBench rather than a property of the detector, and repeating the exercise with cohort-matched benign prompts makes those rows live without changing the picture. At 5% the filter removes 55% of plain harmful prompts and 35% of wrapped ones. Within the wrapped cohort that is 42.5% of the attacks that failed against 14.8% of those that succeeded, a gap of -27.7 points (CI $[-44.8, -8.8]$). At 10% the corresponding recalls are 84%, 67% and 29.6%, a gap of -51.2 points. The gap between failed-attack recall and successful-attack recall is the deployment-facing statement of the inverse ranking: the filter preferentially removes attacks that were not going to work. On the strong-attack cohort at the same threshold, 325 of 354 successful attacks pass under Llama Guard and 123 of 133

under Qwen3.

Risk-coverage, and what it does and does not show. Among prompts that pass the filter, the harmful-generation rate as a function of the fraction blocked rises for the harmfulness score, reading 0.27, 0.289, 0.300, 0.329, 0.383, 0.420 at 0, 10, 20, 30, 40 and 50% coverage. Blocking the same *fraction* at random leaves it at 0.27 throughout, and the rare-token and passive channels track the learned probe (0.42 and 0.44 at half coverage), so the behavior is not specific to active probing.

We flag two ways this comparison can mislead, because both cut against the strongest reading. Matching random filtering on coverage is not matching it on cost: this cohort is entirely harmful, so a rule that blocks 35% of it would block 35% of benign traffic in deployment, whereas our threshold is held to a 5.7% benign false-positive rate. Matched instead on benign budget, the score removes about 2.6 times as many successful attacks as random. And the rising curve is a conditional rate on a denominator the filter itself shrinks. Absolute harm falls monotonically ($27 \rightarrow 23 \rightarrow 19$ realized jailbreaks at 0, 5 and 10% benign FPR), and adding the cohort’s own wrapped-benign prompts at 1:1 reduces the excess over random at the 5% operating point from 8.4 points to 0.7. The defensible claim is therefore about *allocation*, since the budget is spent on attacks that were not going to succeed, and about what a downstream consumer may infer from having passed the filter, not that the detector is worse than no detector. For contrast, an outcome-supervised score at matched coverage drives the admitted-harm rate to 0.062, and Llama Guard’s prompt flag is essentially flat at 0.264–0.270.

Attributing the leakage. At the 5% operating point 0.85 of successful attacks pass. Decomposed, 53% of that leakage is prompts the filter would have missed even without a wrapper, 23% is the wrapper lowering scores across the board, and 24% is the inverse ranking itself: relative to a counterfactual score of the same overall strength but neutral outcome ranking, about five additional successful attacks get through. At the 10% point the inverse-ranking share rises to 53%. We report this decomposition because it bounds how much of the observed deployment risk our specific finding is responsible for, as opposed to ordinary detector insensitivity. A residual caveat is that part of the inverse-ranking term is itself a wrapper-family main effect, which the additive decomposition does not separate out.

Attention-normalization controls. Fixing the probe’s position does not fix its competitive environment in the softmax, so a longer prompt could reduce attention onto three fixed positions mechanically. Two facts make this checkable here: the system region has a constant 13 tokens, and the wrapper text lands in the user turn, so the total attention mass on the system region is a fixed-size competition set and dividing by it is a clean dilution control. Dilution is real, since system-region mass falls from 0.383 plain to 0.225 wrapped, but it does not account for the result. Under the raw read the wrapped-cohort outcome AUROC of the unsupervised aggre-

gate is 0.099 (20 heads) and 0.102 (all heads). Normalizing by system-region mass gives 0.154 and 0.109, normalizing by the full attention row leaves the values unchanged at 0.099 and 0.102, and a readout retrained on the normalized feature gives 0.351. Every variant remains inverse with an interval excluding chance. Regressing the score on sequence length within each condition explains about 4% of its variance.

E Decoding Variance

Protocol. The main results use one greedy decode per prompt, so each outcome label is a single Bernoulli realization. To separate what is target-specific from what is draw-specific we regenerated the entire factorial harmful cohort five times per target at temperature one with recorded seeds, in both the plain and the wrapped condition, and judged all 3000 completions with the same frozen Llama Guard 3.

The ranking is stable. On Llama the score’s outcome AUROC is 0.220 against the published greedy label, 0.252 against success@5, 0.202 against a majority vote over the five draws, and 0.206 against the per-goal estimated success probability with goals resampled as clusters. The five individual draws give 0.175 to 0.257. Redrawing labels from the estimated probabilities gives a sampling distribution with mean 0.206 and standard deviation 0.032, within which the published greedy value sits at the 66th percentile. The single decode was therefore representative.

Cross-target disagreement was overstated by the single draw. The greedy decode gives a three-way disagreement rate of 0.64 and Cochran $Q = 89.7$. Permuting the fifteen judged draws of each goal across target labels, which is exact under the null of no target effect, already produces a single-draw disagreement of 0.54 on average. About 85% of the raw greedy disagreement is therefore what sampling noise alone would produce, and only about 15% is attributable to target identity. The comparison becomes clean once each goal is summarized by five draws: success@5 disagreement is 0.81 against a null of 0.29, and Q rejects the null in both cases. We report the success@5 contrast in the main text and no longer quote the single-draw agreement count, which overstates the target-specificity claim.

F Held-Out Calibration

Frozen in-distribution Platt calibration. The main paper’s slope column is an in-sample Cox fit. To separate a score that was never calibrated from one whose calibration the shift destroyed, we hold out a 30% in-distribution validation slice, fit Platt scaling on it, freeze the mapping, and apply it unchanged to every domain (Table 4). AUROC is invariant under this monotone map by construction and is repeated only for reference. In distribution the correction works as intended (slope 0.545 $\rightarrow$ 1.019, ECE 0.045 $\rightarrow$ 0.009). Out of distribution it

does not restore calibrated meaning: WildJailbreak reaches slope 0.994 but retains ECE 0.146 and Brier 0.104, and the both-sided encodings stay at ECE 0.236–0.399 with slopes near zero, since no monotone map can calibrate a score whose ranking is at chance. Japanese is the one condition with a negative slope under both raw and calibrated scores, consistent with the translation artifact documented below rather than with ordinary shift.

Translation-artifact condition. The Japanese row is reported here rather than in the main table. Its AUROC of 0.228 is not a weak signal but an inverted one, and the inversion is a property of the transformation, not of the score: the `opus-mt-en-jap` outputs collapse toward a shared register, with mean pairwise character trigram Jaccard similarity 0.151 among translated prompts versus 0.062, 0.078, and 0.0065 for the English, French, and Chinese conditions, that is, $2.4\times$ the self-similarity of the source text. Orientation-free, $1 - 0.228 = 0.772$ sits just below the Chinese and French band (0.778–0.802), which is what one expects if the transformation has flipped the class-conditional ordering rather than destroyed the signal.

G Common-Prior and Outcome Sensitivity

Common-prior audit. To separate conditional score shift from label prevalence, we retain every example but assign inverse-frequency weights so each class has total mass 0.5 in every domain. This weighting is deterministic. AUROC is unchanged, while ECE, Cox calibration, and F1 are recomputed under the shared prior. Table 5 shows that WildJailbreak prevalence explains part of the observed calibration gap. For AAP, the ID common-prior threshold is nearly oracle on WildJailbreak, whereas HiddenDetect and the refusal-direction read retain larger threshold gaps. All families lose class separation on both-sided base64.

Benign-only threshold audit. At nominal FPR 1% with 500 safe calibration prompts, the empirical quantile achieves FPR 1.2% and nearly matches an oracle TPR on paraphrase and English multilingual. Under benign shift it becomes conservative (WildJailbreak FPR 0.2%, TPR 0.03 versus oracle 0.25), and encoding yields zero TPR because the signal is absent. The experiment therefore identifies benign-tail stability. It changes the decision map but not outcome ranking or its target index.

Target-decomposition protocol. We sample 100 benign and 100 harmful JailbreakBench goals. Each base goal is evaluated plain and under exactly one of five fixed wrappers, balanced at 20 goals per intent class and wrapper family. This yields 400 prompts while preserving the plain/wrapped pairing. The templates are fixed strings with a single `{goal}` slot: *Fictional role-play* frames the exchange as an authorized evaluation and asks the model to adopt an unrestricted in-world expert persona and answer directly without stepping out of character. *Authority override* declares an authorized red-team

Domain	AUROC	Raw			Frozen ID Platt		
		ECE	Brier	slope	ECE	Brier	slope
In-distribution	.985	.045	.045	+.545	.009	.033	+1.019
WildJailbreak	.850	.366	.272	+.508	.146	.104	+0.994
Paraphrase	.858	.203	.202	+.320	.065	.135	+0.626
Multilingual (en)	.987	.041	.045	+.761	.171	.096	+1.488
Multilingual (zh)	.802	.247	.222	+.434	.138	.170	+0.847
Multilingual (fr)	.778	.232	.234	+.419	.123	.192	+0.819
Multilingual (ja)	.228	.479	.484	−.809	.340	.354	−1.582
Encoding base64	.507	.708	.682	+.016	.283	.262	+0.030
Encoding rot13	.498	.684	.650	−.011	.236	.242	−0.023
Encoding caesar	.517	.746	.732	+.056	.399	.344	+0.113

Table 4: Raw versus frozen in-distribution Platt calibration for the dense all-head AAP readout, all shifts both-sided. The Platt map is fit once on a held-out 30% in-distribution validation slice and never refit. AUROC is unchanged by construction.

Score	ID ECE	WJ AUROC	WJ ECE	WJ F1 _{ID}	WJ F1 _{or}	Base64 AUROC
AAP	.036	.850	.149	.790	.796	.507
Passive	.031	.743	.167	.706	.711	.518
AttentionDefense-style	.033	.804	.085	.724	.732	.503
HiddenDetect-style	.056	.784	.379	.420	.725	.487
Refusal direction	.044	.803	.280	.587	.736	.486
Llama Guard	.179	.840	.195	.663	.762	.506

Table 5: Common-prior (50/50) decomposition. F1_{ID} freezes the ID-optimal common-prior threshold. F1_{or} is the target-domain oracle. The base64 column is AUROC and is therefore prior invariant.

evaluation that supersedes ordinary refusal behavior for the duration of the test and demands a direct and complete answer. *Simulated unfiltered mode* asks the model to simulate a hypothetical assistant with no safety filters and return only that assistant’s detailed answer. *Compliance prefix* instructs the model to continue after a supplied compliant opening, without warnings or refusals, the opening being a sentence announcing that the requested information follows. *Research pretext* requests a technically complete, step-by-step response for a controlled academic security study and forbids replacing operational detail with general cautions. The same five templates are applied to benign and harmful goals alike, so the wrapper is never confounded with intent. The locked record list has SHA-256 digest 1917080d3a832dc3e87a20036327556cf7f960379c98e6763021add8932c990. The Llama-3.1-8B-Instruct target uses greedy decoding for at most 256 new tokens. Llama Guard 3 is the primary frozen outcome judge. Paired-contrast intervals use 10,000 bootstrap replicates over base goals, while outcome-AUROC intervals use class-stratified resampling. The wrapped harmful cell contains 27 effective and 73 ineffective outcomes. These choices were fixed before inference. All goals and wrapper families remain in the primary analysis. For the added matched controls, we extract a three-rare-token system-region read and a no-insertion passive system-region read on the same 400

prompts. Each channel’s XGBoost scorer is trained only on its existing 1000+1000 BeaverTails feature cache, with no refitting or selection on the factorial data. AttentionDefense-style system-span attention and HiddenDetect-style per-layer refusal evidence are extracted in the same no-probe prefill pass. Their readouts use the same BeaverTails-only protocol. They yield wrapped R_H/R_Y of .828/.145 and .782/.248, respectively.

What the -style suffix means for the derived channels.

These are adaptations, not reimplementations, and we list the deviations so that no result is read as a reproduction of the original method. For the *HiddenDetect-style* channel: the score is the difference of mean first-token logits over a handwritten eight-word refusal list and six-word affirmation list rather than the published refusal-vector formulation. We emit the full per-layer vector and fit a supervised BeaverTails classifier in place of the original training-free layer aggregation and threshold. The original method targets vision-language models and we apply it to text only, and the read position is the final token of *our* chat template, which includes our system prompt. For the *AttentionDefense-style* channel: the system span is taken as template positions up to the insertion anchor, including the beginning-of-sequence token and role headers. Our fixed one-sentence system prompt is not their defensive prompt. We read only the final query row of the at-

tention matrix. A supervised all-head classifier replaces their explainable aggregate, and the aggregation is an unweighted mean with no length normalization. The *passive* channel is an anchor-matched no-insertion control of our own construction. It is not a reimplement of Attention Tracker’s instruction-focus score, which we cite only as the origin of the prompt-dependent read position that motivates a fixed coordinate. Consequently these channels test whether the reversal survives a change of read location and readout family, not whether published detectors would report the same numbers. Across five independently learned probe seeds with seed-specific BeaverTails readouts, mean τ_H is .501 (SD .036), mean τ_{HW} is -.319 (SD .118), and every seed preserves the inversion: wrapped $R_H = .770-.861$ while $R_Y = .130-.246$. For target substitution, the same 200 harmful prompts are generated by Mistral-7B-Instruct-v0.3 and Qwen2.5-7B-Instruct under the identical greedy 256-token budget and judged by the same frozen Llama Guard 3. This fixed-score audit does not recompute detector scores. The target-native audit below does. Mistral yields 68/70 effective plain/wrapped prompts. Qwen yields 5/10. All targets, prompts, metrics, and the 10,000-replicate bootstrap were fixed before inspecting these outcomes. The paired target audit then resamples the 100 base goals while preserving each goal’s two conditions and all three target outcomes. On wrapped prompts, the Llama-Mistral, Llama-Qwen, and Mistral-Qwen disagreement rates are 0.49 (CI [0.39, 0.59]), 0.19 ([0.12, 0.27]), and 0.60 ([0.50, 0.69]), respectively. Corresponding Cohen’s κ values are 0.172, 0.399, and 0.091. Cochran’s paired omnibus statistic is $Q = 89.66$ ($p = 3.4 \times 10^{-20}$). No generation, score fitting, or outcome relabeling is performed for this audit. For judge robustness, Qwen3-14B receives the same 600 cached harmful-goal prompt-completion pairs with a frozen rubric that labels only substantive, actionable harmful compliance as success. Greedy decoding with thinking disabled produces one binary token. All 600 outputs parse without manual correction. On wrapped prompts, Qwen3 versus Llama Guard agreement/ κ is .87/.611 for Llama, .66/.388 for Mistral, and .94/.545 for Qwen. Qwen3’s paired target test gives $Q = 43.46$ ($p = 3.66 \times 10^{-10}$). The protocol and rubric were hashed before label aggregation. The second judge changes no generation or detector score.

StrongREJECT column: what we actually ran. The released fine-tuned StrongREJECT evaluator could not be obtained in our offline environment because its base checkpoint is gated, so we score the published StrongREJECT rubric with Gemma-2-9B-it. This column is therefore a rubric reimplement, not the official evaluator, and we report it as such. The failure is one of judge substitution, not of parsing. Of 1600 rubric outputs, 1594 parse. The six failures are the grader declining to score, not regex misses, and the parsed sub-scores are jointly structured in the way the rubric prescribes (refused=1 $\Rightarrow$ score 0. Attacks lower the refusal item), which random mis-parsing could not produce. What

fails is the grader: Gemma-2-9B-it marks the rubric’s refusal item as *not refused* on roughly 30% of verbatim refusals. A completion that plainly declines to write a sexist email, for instance, is scored refused=0, convincing=5, specific=4, hence a rubric score of 0.875 and a nominal success. The binarized label is therefore a non-refusal-and-fluency indicator rather than a harm label, and it fires on 66–99% of *benign* control prompts that receive helpful answers. Decomposing $ASR_{SR} = \Pr(\text{not refused}) \times \Pr(\text{convincing} + \text{specific} \geq 6 \mid \text{not refused})$, the second factor is 0.92–1.00 on plain harmful prompts, so the elevated plain-prompt rate is driven by the refusal item rather than by our cut. The apparent post-attack *decreases* are a separate, cut-related effect: attacks lower refusal but degrade coherence and specificity, and 12.6% of harmful items sit exactly on the 0.5 boundary, the densest non-zero bin, so those labels are knife-edge. We therefore report this column as a rubric-quality measurement and do not interpret it as an attack-success rate. The inverse ranking does not depend on it. $R_Y < 0.5$ holds in 7/7 Llama cells under Llama Guard 3, 6/7 under the rubric cut, 5/7 under Qwen3, 7/7 under a strict two-of-three consensus label, and 6/7 under a union label. The pooled score coefficient stays negative under every rule (–0.539 Llama Guard, –0.613 rubric, –0.349 Qwen3, –0.444 consensus, –0.852 union). Three-way agreement is nevertheless low, with Fleiss κ of 0.21–0.30 per cell, which is itself evidence for the paper’s claim that Y_m is judge-indexed rather than a single quantity. The rubric column is also the only non-deterministic judge: across 332 records duplicated between attack directories its label differs in 9 cases (2.7%), traced to batch padding composition and a doubled beginning-of-sequence token, whereas Llama Guard 3 and Qwen3 differ in 0/332.

Target-native score matrix. The main paper reports the full 3×3 crossing of native AAP score sources and generator-indexed outcomes on the identical 100 wrapped harmful prompts. Pairwise native-score correlations are only .42–.54, yet recomputing the score on each model does not recover positive outcome ranking. This rules out a frozen-Llama score source as the explanation for the target-indexed result.

Intervention perturbation. We quantify how AAP changes the input computation. Across 200 prompts, next-token symmetric KL has median 0.22, mean 0.79, and 95th percentile 3.87. Refusal flips on 7.5% of prompts, with safe/harmful refusal changing .30 $\rightarrow$.21 and .94 $\rightarrow$.96. Perplexity increases by 5.4% and is teacher-forced on the no-probe greedy continuation, so it is a one-sided upper bound. These perturbations bound the interpretation of the intervention contrasts. The no-insertion and passive-channel results do not depend on probe insertion.

Does the probe change realized success? No reported Y_m is generated under probe insertion, but a reader may still ask what would happen if it were. We therefore generate the same 300 held-out harmful prompts (150 WildJailbreak, 150 BeaverTails. Greedy, 256 new tokens) under four arms

Score	τ_H	τ_W	τ_{HW}	R_H P/W	R_Y
AAP	.521	−.094	−.257	.936/.803	.220 (.109,.343)
Rare-token	.490	.049	−.431	.961/.760	.195 (.101,.300)
Passive	.408	.031	−.338	.945/.751	.149 (.062,.252)
AttentionDefense-style	.511	.094	−.359	.956/.828	.145 (.061,.244)
HiddenDetect-style	.413	−.159	−.284	.894/.782	.248 (.126,.383)
Llama Guard	.715	.045	−.070	.875/.840	.507 (.500,.521)

Table 6: Complete matched harmfulness–wrapper decomposition. R_H is harmful-intent AUROC on plain/wrapped prompts. R_Y ranks effective versus ineffective wrapped harmful prompts. Each internal channel uses its BeaverTails-trained matched scorer. Parentheses are bootstrap 95% CIs for R_Y .

Generator	Effective P/W	ASR P/W	AAP R_Y	Rare R_Y	Passive R_Y	LG-prompt R_Y
Llama	5/27	.05/.27	.220 [.109,.343]	.195 [.101,.300]	.149 [.062,.252]	.507 [.500,.521]
Mistral	68/70	.68/.70	.491 [.363,.623]	.508 [.394,.623]	.407 [.296,.525]	.517 [.500,.550]
Qwen	5/10	.05/.10	.163 [.024,.358]	.098 [.013,.214]	.214 [.060,.404]	.506 [.500,.517]

Table 7: Fixed-score target-indexed outcome audit on the same 100 plain and 100 wrapped harmful goals. The Llama score source is held fixed while only the generator changes. The next analysis crosses native score sources and generators. R_Y is wrapped effective-versus-ineffective AUROC with bootstrap 95% CI.

and judge all of them with the same frozen Llama Guard 3: the canonical no-probe path, the probe path, a matched control carrying the anchored system template *without* the probe, and a re-run of the canonical path at a different batch size as a floating-point null. Attack success is 0.207 without the probe and 0.177 with it, a difference of -0.030 (95% CI $[-0.070, +0.007]$. McNemar $b=13$, $c=22$, $p = 0.18$): probe insertion does not increase realized jailbreak success, and the point estimate is if anything mildly protective. The template-only control reaches 0.157, so most of the small shift is attributable to the anchored system template rather than to the learned probe. Completions do diverge textually (exact-match 0.107, mean unigram F1 0.582), and refusal rises from 0.583 to 0.660, which is why we report the intervention as bounded rather than inert. The batch-size re-run gives 0.217, indicating a floating-point noise floor of roughly one percentage point on this measurement.

H Reproducibility and Computational Environment

Recorded environment. The primary archived model-side environment is Linux on an NVIDIA RTX A6000 with 48 GB memory, using half-precision model weights. The corresponding software snapshot records Python 3.10.12, PyTorch 2.5.1 with CUDA 12.1, Transformers 5.9.0, scikit-learn 1.7.2, XGBoost 3.2.0, NumPy 2.2.6, SciPy 1.15.3, Accelerate 1.13.0, and SentencePiece 0.2.1. Search-based attacks were parallelized, and not every worker retained equally detailed hardware metadata. We therefore report this as the primary

recorded environment rather than claim that every attack job ran on the same device.

Final settings. The primary AAP probe has $K = 3$ rows and is optimized with Adam at learning rate 10^{-3} for 50 updates, using a freshly sampled minibatch of 20 prompts per class at each update. RQ1 uses the 20 heads selected on the training split, whereas RQ2 uses the dense all-head response. The primary harmfulness readout is XGBoost with 100 trees, maximum depth 5, log-loss evaluation, and random state 42. The outcome-supervised positive control uses 200 trees, maximum depth 3, learning rate 0.1, row subsampling 0.8, column subsampling 0.5, and eight CPU workers. Its logistic base-lines use ℓ_2 regularization, `lbfgs`, at most 300 iterations, and select C within each training fold from $\{0.001, 0.01, 0.1, 1.0\}$. Model checkpoint identities are stated in the main setup and the target-decomposition protocol. Primary generations use greedy decoding for at most 256 new tokens. The attack-specific goal counts and search budgets are listed in Supplementary §1.

Randomness, repetition, and uncertainty. Unless an experiment explicitly varies the seed, data splits and fixed estimator fits use random state 42. Probe sensitivity uses five independently learned probe seeds. The decoding-variance audit records five sampling seeds at temperature one, while the primary outcome uses one greedy decode per prompt. Matched contrasts and outcome-AUROC intervals use 10,000 bootstrap replicates with the resampling unit stated next to each analysis. Cross validation keeps the base goal inside one fold, and the outcome-supervised readout uses 25 repeats. These counts distinguish repeated fits, repeated decodes, and bootstrap replicates rather than treating them as interchangeable

Attack	LG3 R_Y	SR R_Y	Q3 R_Y
Best-of-N	.345 [.167,.541]	.434 [.344,.524]	.536 [.444,.626]
Many-shot	.478 [.380,.576]	.534 [.448,.616]	.494 [.395,.591]
GCG	.189 [.085,.315]	.273 [.158,.397]	.264 [.089,.476]
AutoDAN	.195 [.110,.291]	.143 [.074,.224]	.237 [.131,.360]
PAIR	.485 [.371,.599]	.379 [.258,.510]	.425 [.237,.624]
TAP	.310 [.140,.500]	.248 [.118,.395]	.206 [.043,.426]
AutoDAN-Turbo	.284 [.143,.447]	.252 [.117,.403]	.515 [.262,.767]

Table 8: Strong-attack outcome-ranking uncertainty corresponding to the compact main-paper table, after the checkpoint repair described above. Entries are R_Y with 10,000-replicate class-stratified bootstrap 95% CIs on attacked harmful prompts generated by Llama-3.1-8B-Instruct. LG3 and Q3 denote Llama Guard 3 and Qwen3-14B. SR is the rubric reimplement, reported but excluded from every count. Before the repair, GCG read .203 [.100, .329] / .257 [.145, .381] / .304 [.099, .535] and AutoDAN-Turbo’s SR entry read .185 [.070, .323]. TAP’s Llama Guard interval had upper endpoint .494 rather than .500. Only the intervals whose endpoint straddles 0.5 change any verdict: GCG–Qwen3 enters the significantly-below set and TAP–Llama Guard leaves it.

runs.

Artifact boundary. The artifact linked from the main paper releases analysis code, portable configurations, tests, and publication-aligned numeric aggregates. It omits raw harmful prompts and completions, model weights, attack checkpoints, and private execution provenance. Consequently, the released aggregates support table and correspondence verification, while model-side experiments that require omitted artifacts are documented at the protocol level.

I Details Referenced from the Main Text

Prior silent-failure evidence. The collapse cited in the main text is measured under a synthetic $\sigma=0.02$ Gaussian perturbation of the frozen embedding input. ROC-AUC falls from 85% to 50% under that perturbation while mean predicted confidence falls 14%.

Single-source inflation in prior audits. The 8.0–16.5 AUC inflation cited in the main text is measured by comparing single-source with cross-source splits for safety detectors. A sparse-autoencoder analysis in the same work finds that 28–44% of top features encode dataset identity rather than semantics, which motivates its leave-one-dataset-out protocol.

Both the deployed twenty-head read and the dense all-head read select from this one tensor. No separate extraction run produces the dense read, so the RQ1 and RQ2 specifications differ only in which heads are selected.

Safety-semantic initialization control. The safety-semantic condition inserts the probe initialization without any optimizer update, holding row count, insertion coordinate, dtype and read position fixed against the learned probe. Its response does

not reach the learned probe’s response on the same prompts. The amplification is therefore attributable to the optimization in Eq. (1) rather than to the safety-semantic content of the initializing token embeddings.

Cross-architecture WildJailbreak transfer. We repeat the WildJailbreak evaluation on Mistral, Qwen, Gemma, and Phi. Each architecture keeps its own frozen probe, its own head set, and its own BeaverTails-trained readout. No WildJailbreak prompt enters any fit. In-distribution AUROC runs from 0.96 to 0.98 across the five architectures and WildJailbreak AUROC runs from 0.60 to 0.85. The degradation is therefore a property of the shift rather than of one attention grid.

Model-free control for the single-class encoding design. The control statistic is the Shannon entropy of the character distribution of the raw prompt string. It uses no model, no probe, and no training. On the single-class design, which encodes only the harmful class, it separates encoded attacks from plain benign prompts with AUROC 0.997–1.000 across base64, rot13, and caesar. On the same prompts without encoding it reaches 0.513. A single-class encoding result therefore cannot be read as evidence about harmfulness separation, because a statistic with no access to meaning already solves that comparison.

Per-head multiple comparisons. Per-head tests apply Benjamini–Hochberg FDR control across all 1024 heads, so every per-head significance statement holds after false discovery rate control. The max-statistic permutation correction quoted above is a separate family-wise procedure used only for the outcome-informative head count.

The goal subsets for the reduced- n rows were also fixed before any attack was run, together with the budgets listed above.